Class 3:

1. Find the Fourier transform of $f(t) = te^{-|t|}$ Ans: $\frac{4i\omega}{\left(\omega^2+1\right)^2}$

2. Find the Fourier transform of $f(t) = \begin{cases} \frac{\pi}{2}\cos t & for\ |t| \le \pi \\ 0 & for\ |t| > \pi \end{cases}$. Ans: $\frac{\pi\omega(-1)^n \sin\pi\omega}{\omega^2-1}$

3. Find the Fourier transform of $f(t) = e^{-a^2x^2}$ and hence show that $e^{-\frac{x^2}{2}}$ is self reciprocal.

Ans: $\frac{\sqrt{\pi}}{a}e^{-\frac{\omega^2}{4a^2}}$

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